

TCTrack: FAIR Tropical Cyclone Tracking

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Summary

TCTrack is a Python package providing a unified, user-friendly interface to multiple established tropical cyclone tracking algorithms: TRACK ([Hodges et al., 2017](#)), TempestExtremes ([Ullrich et al., 2021](#)), and TSTORMS ([Vitart et al., 1997](#)). Whilst these algorithms are widely used in the climate science community, each has its own unique interface, input requirements, and output format, making intercomparison studies and routine application challenging. TCTrack addresses these usability issues by wrapping each algorithm with a consistent Python API based on abstract base classes. Crucially, TCTrack transforms the output from each algorithm into a standardised NetCDF format that is fully compliant with the Climate and Forecasting (CF) conventions ([Earth Science Data Systems, 2024](#)) using the discrete sampling geometry trajectory format, ensuring that all metadata from input data and processing parameters are preserved. This approach makes TCTrack particularly valuable for climate model evaluation studies, such as those in CMIP (the Coupled Model Intercomparison Project), and for tracking algorithm intercomparison work. By providing standardised, metadata-rich outputs, TCTrack aligns with the FAIR principles for research data and software ([Barker et al., 2022](#); [Wilkinson et al., 2016](#)), ensuring outputs are Findable, Accessible, Interoperable, and Reusable.

Statement of need

Tropical cyclone tracking is an essential component of climate model evaluation and climate change impact assessment. However, researchers face significant barriers when attempting to use existing tracking codes. Each established algorithm—TRACK, TempestExtremes, and TSTORMS—requires separate installation, has distinct input data requirements and parameter specifications, and produces output in custom formats with limited or no metadata. Users who wish to compare results across multiple tracking algorithms must invest substantial effort learning each tool independently, converting between different data formats, and manually adding provenance metadata to outputs.

TCTrack streamlines this workflow by providing a uniform interface across all three algorithms. Users need only:

- 1) configure tracking parameters using algorithm-specific parameter dataclasses,
- 2) instantiate a tracker object and call the `run_tracker()` method,
- 3) use the CF-compliant netCDF output data.

All output files follow the CF conventions' discrete sampling geometry trajectory format (sections 9 and H4 of CF-1.13 ([Earth Science Data Systems, 2024](#))), with coordinate variables, auxiliary coordinates, and complete metadata including algorithm name, TCTrack version, and all parameters required for reproducibility.

This standardised framework that TCTrack provides will support researchers, forecasters, and the natural hazard industry (e.g. insurers) for using different algorithms to project the impact of tropical cyclones. It is particularly valuable for two key use cases. First, climate model evaluation studies—such as the assessment of tropical cyclones in Neural GCM ([Kochkov et](#)

42 [al., 2024](#))—require consistent application of tracking algorithms to model outputs. Second,
43 intercomparison studies are essential given documented variations in cyclone statistics between
44 different CMIP models ([Bourdin et al., 2024](#)), tracking methods ([Roberts et al., 2020](#)), and
45 reanalysis datasets ([Hodges et al., 2017](#)). TCTrack facilitates such studies by providing a
46 common framework whilst preserving the distinct characteristics of each underlying algorithm.

47 Software Design

48 TCTrack is implemented as a Python package built using object-oriented design. Users select
49 an appropriate parameter class to configure tracking, then pass this to a tracker class in
50 order to run and output results. These use abstract base classes to present a standardised
51 API across the different tracking algorithms and clear interfaces for adding new algorithms.
52 Meanwhile, the algorithm-specific details are abstracted away from the end user by the concrete
53 implementations of these classes.

54 It is possible for tracking algorithms to be implemented directly in TCTrack, but when building
55 on existing tracking codes we utilise subprocess calls to reduce the software maintenance
56 required and give users access to the proven underlying programs.

57 A particularly important aspect of TCTrack is the standardised, CF-compliant output format
58 into which the output tracks are transformed; achieved using the cf-python software package
59 ([Hassell et al., 2017](#); [Hassell & Bartholomew, 2020](#)). This ensures that outputs are FAIR
60 (which is of increasing importance in climate science) and immediately usable in downstream
61 analysis, something that is not a feature of existing approaches. Metadata is added to the
62 output by capturing the metadata from input files and augmenting it to reflect any processing
63 of variables, such as spatial aggregation. It also provides the tracking algorithm used, TCTrack
64 version, and complete parameter specifications for reproducibility.

65 The codebase includes comprehensive unit tests using pytest and continuous integration via
66 GitHub Actions to ensure validity and compatibility across multiple Python versions. Code
67 quality is maintained through static analysis to provide robustness and maintainability.

68 The package is available via GitHub at <https://github.com/Cambridge-ICCS/TCTrack> and
69 extensively documented at <https://tctrack.readthedocs.io/>. This includes installation guidance,
70 API reference, and worked examples.

71 Research Impact Statement

72 TCTrack addresses a need within the community for simple, accessible tooling for exploring the
73 differences in tropical cyclone tracks and statistics produced by different tracking algorithms
74 and conducting intercomparison studies. It has been used in climate science research as part of
75 CMIP6 model comparison studies by researchers at the University of Cambridge, with results
76 presented to the scientific community in [Atkinson et al. \(2026\)](#). This work also aims to
77 reproduce and update the findings of [Knutson et al. \(2020\)](#) to project the response of tropical
78 cyclone activity to future climate change using more recent CMIP6 data. TCTrack is also
79 being used in industry by INIGO Insurance to generate tracks in a robust, standardised format
80 for downstream use in data analysis and presentation.

81 State of the Field

82 The three tracking algorithms provided by TCTrack (TRACK, TempestExtremes, and
83 TSTORMS) are established, peer-reviewed methods used in the climate science community.
84 TRACK uses a feature-tracking approach based on vorticity maxima and has been applied to
85 assess tropical cyclones in reanalysis datasets ([Hodges et al., 2017](#)), TempestExtremes employs
86 geometric criteria on multiple variables ([Ullrich et al., 2021](#)), and TSTORMS searches for

co-located vorticity and temperature maxima and sea-level pressure minima (Vitart et al., 1997). However, their native interfaces are command-line based with algorithm-specific input formats and minimal output metadata, creating barriers to adoption and intercomparison.

Alternative Python-based tracking tools exist, such as tobac (Heikenfeld et al., 2019), which provides general feature tracking capabilities primarily for meteorological applications. However, tobac implements new tracking methodologies rather than providing access to the established algorithms that have been extensively used and validated in the climate community. TCTrack's approach differs fundamentally: it enables researchers to use proven methods whilst addressing the practical challenges of usability, interoperability, and metadata standards. TCTrack is also extensible in that additional tracking algorithms can be integrated following the contributor guidelines and building on the abstract base class design.

The decision to build TCTrack rather than change the data format/running procedures of multiple existing projects comes from the desire to also provide a unified interface for researchers as well as improved data representation. TCTrack's focus on usability and FAIR data builds on what the existing tools offer, making it a valuable part of the ecosystem.

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AI Usage Disclosure

During the development of this software AI tools (ChatGPT, Devstral) were used to understand usage of functions, review written code to suggest issues and improvements, and for generating boilerplate and parameterisation of tests. Any code suggestions by AI were thoroughly checked and double checked during code review by a second person. During the preparation of this paper, AI was used to refactor an initial draft to align with the changes to the JOSS scope. This was thoroughly checked and built upon thereafter.

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